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ENGR095 - Sensing the World

Reflection 1

The first week of the Sensing the World module of the Engineering 095 FYRE program began with an overview of material science and the science behind microchips. Our class got a tour of the clean room where microchips are made, and we studied the process of making microchips. I enjoyed learning about how photosensitive chemicals and light masks are implemented to make specific circuit designs. We also got a tour of the Wilbur Powerhouse and were shown how to use the vinyl cutter, laser cutter, 3d printers, and circuit printers. We met our student advisors, who showed us various projects which would be completed in the powerhouse. We were also given access to the website that allows us to get certified to use various machines in Wilbur powerhouse. During our lab periods, we cut out IDE's (Inter Digitated Electrodes) and spin-coated them with various conductive materials. We then tested conductivity of different materials and explored how they reacted to different conditions. I learned that materials can drastically change their conductivity just by being in contact with water vapor. In class, we created LED circuits on paper with copper tape. I learned how contact switches work, what polarized components are, and how to make a parallel circuit.

Most recently, we created our own GitHub profile. We used copilot to create code for our websites which we will use to display all of our work, similar to a resume. We also connected our computers to Arduino Nano's which we coded with some simple code. We then added our codes to our README on GitHub and added a link to GitHub to our portfolio. So far, my experience has been very intriguing and fulfilling. I've learned so much about many different disciplines of engineering and engineering processes. I've gained a new interest in materials engineering and materials science which I wouldn't have guessed that I'd be interested in. I'm excited to continue learning about how microchips have evolved over time and how AI is impacting the world of programming and computer science. I'm super glad that I got into the FYRE program and it's making me feel more dedicated to continuing my engineering career, even if some parts are difficult.